

ω Cen: orbital histories of the progenitor dwarf

Three classes of initial conditions for the host-disruption simulations

oCen_dm project notes

2026 September 20

Abstract

The mass-modelling track infers ω Cen’s dark-matter content from its present-day kinematics. This note prepares the complementary track: simulate the tidal disruption of the host dwarf and ask how much dark matter the surviving nucleus is left with. A disruption simulation needs an orbital history, and three classes were defined: (1) the present orbit in a static axisymmetric potential; (2) the present orbit integrated backwards with dynamical friction, so the dwarf arrives from the outer halo and sinks; (3) deposition by the Gaia–Sausage–Enceladus (GSE) merger followed by inward migration driven by the decelerating Galactic bar (Dillamore et al., 2026). For each class we state the model, examine the assumptions that matter, and pick three representative orbits; the nine orbits are then compared through per-Gyr pericentres, apocentres, tidal-field strength and Jacobi radii. The main findings: class 2 is controlled by the satellite’s mass-loss history rather than its mass, and every viable history converges to today’s orbit over the last 3–5 Gyr; class 3 reproduces the published result that migration from GSE debris requires a present-day pattern speed $\Omega_b \lesssim 26 \text{ km s}^{-1} \text{ kpc}^{-1}$, places the nucleus on a most-bound GSE-debris orbit (apocentre 11–12 kpc, pericentre 0.4 kpc) until ~ 2.5 Gyr ago, and shows that a slow bar also alters “today’s orbit” (pericentre 0.8 rather than 1.6 kpc). The classes differ most in the pericentric tides 5–10 Gyr ago, by more than an order of magnitude.

1 Purpose and scope

The question of the project is whether ω Cen is embedded in dark matter. The kinematic route constrains the mass profile today. The dynamical route asks what a disrupting dwarf leaves behind, which depends on how the dwarf was stripped, hence on its orbit through the Milky Way. Before running any N -body model we fix a small set of orbital histories that span the plausible range. This note is about those histories only; no disruption has been simulated.

All code is in `src/ocen_dm/tails/`: `progenitor_orbits.py` (classes 1 and 2, the orbit set), `bar_migration.py` (class 3), `products.py` and `writeup_figures.py` (tables and figures). Tests: `tests/test_progenitor_orbits.py`, `tests/test_bar_migration.py` (11 tests, all passing). Numerical products are in `results/tails/`.

2 Present-day phase-space point

All classes start from the same observables: the centre of `cluster.py` $(\alpha, \delta) = (201.69683^\circ, -47.47957^\circ)$, distance $D = 5.43 \pm 0.05 \text{ kpc}$, systemic proper motion $(\mu_{\alpha*}, \mu_\delta) = (-3.257, -6.730) \pm 0.025 \text{ mas yr}^{-1}$ (Vasiliev & Baumgardt, 2021), line-of-sight velocity $232.7 \pm 0.21 \text{ km s}^{-1}$ (Baumgardt & Vasiliev, 2021). Classes 1 and 2 use the solar parameters of the McMillan (2017) potential, $R_0 = 8.21 \text{ kpc}$, $v_0 = 233.1 \text{ km s}^{-1}$, $z_\odot = 20.8 \text{ pc}$, $(U, V, W)_\odot = (11.1, 12.24, 7.25) \text{ km s}^{-1}$ (Schönrich et al., 2010); class 3 follows Dillamore et al. (2026) and uses the `astropy` default frame ($R_0 = 8.122 \text{ kpc}$, $\mathbf{v}_\odot = (12.9, 245.6, 7.78) \text{ km s}^{-1}$).

Table 1: Present-day galactocentric state (McMillan 2017 frame). L_z is prograde-positive throughout this [note](#).

(x, y, z)	$(-4.90, -4.07, 1.41)$ kpc, $r = 6.52$ kpc
(v_x, v_y, v_z)	$(95.2, -28.9, -87.1)$ km s $^{-1}$
L_z	-529 kpc km s $^{-1}$ (retrograde)
E (McMillan 2017)	-1.85×10^5 km 2 s $^{-2}$
E (Hunter et al. 2024, axisymmetrised)	-1.46×10^5 km 2 s $^{-2}$

A frame error was caught and is now guarded by a test: `galpy`'s `solarmotion` is the peculiar solar motion alone, and adding v_0 to it produced a prograde orbit with apocentre 10.6 kpc. The corrected conversion agrees with an independent `astropy` transform to 0.01 kpc and 0.1 km s $^{-1}$.

3 Class 1: today's orbit in a static axisymmetric potential

3.1 Construction

The present-day state is integrated backwards for 10 Gyr with `galpy` (`dop853_c`, [Bovy, 2015](#)) in three published potentials: McMillan (2017), the `galpy` `MWPotential2014`, and model I of [Irrgang et al. \(2013\)](#). No friction, no mass loss, no time dependence: the dwarf is assumed to have been on this orbit throughout.

3.2 Assumptions

The only assumption is the potential. The three chosen bracket the range of circular-speed curves and bulge/disc masses in common use; the potentials' own solar parameters are used (Irrgang: $R_0 = 8.4$ kpc, $v_0 = 242$ km s $^{-1}$).

3.3 Result

Table 2: Class-1 orbits (kpc). McMillan 2017 reproduces the AGAMA orbit computed on 2026-09-19 for the Jacobi radius (1.59/6.99); the two implementations differ by 4% in pericentre.

potential	r_{peri}	r_{apo}	e	z_{max}
McMillan 2017	1.57	7.04	0.63	2.88
MWPotential2014	1.97	6.76	0.55	2.72
Irrgang 2013 model I	1.25	7.20	0.70	3.06

The pericentre varies by a factor 1.6 between potentials (Fig. 1); the apocentre by 6%. All three are kept: the class-1 spread is the potential spread.

4 Class 2: today's orbit with dynamical friction integrated backwards

4.1 Construction

The equation of motion is integrated backwards in time with the Chandrasekhar deceleration ([Chandrasekhar, 1943](#)) added,

$$\mathbf{a}_{\text{df}} = -4\pi G^2 M(t) \rho(r) \ln \Lambda \left[\text{erf}(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2} \right] \frac{\mathbf{v}}{v^3}, \quad X = \frac{v}{\sqrt{2}\sigma}, \quad \sigma = \frac{v_c(r)}{\sqrt{2}}, \quad \ln \Lambda = \ln \frac{r}{\max(r_h, GM/v^2)} \quad (1)$$

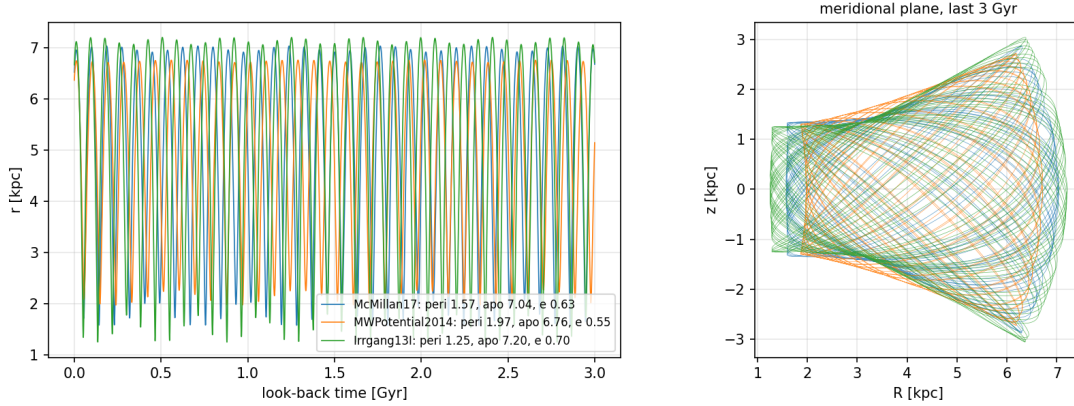


Figure 1: Class 1. Left: galactocentric radius over the last 3 Gyr in the three potentials. Right: the same orbits in the meridional plane. The orbit is a thick, eccentric, retrograde loop confined to $z < 3$ kpc.

which is `galpy`’s prescription with its default Coulomb logarithm. Reversing time reverses the sign of the friction term, so the orbit is pumped outwards: the integration recovers where a satellite of bound mass $M(t)$ must have been to sink to the present orbit. The host is McMillan (2017) evaluated with AGAMA (Vasiliev, 2019); the integrator is a leapfrog with 0.25 Myr steps (about 1 s per 10 Gyr orbit). The frictionless run reproduces class 1 (1.63/7.04 versus 1.57/7.04 kpc), and a constant-mass case checked against `galpy`’s `ChandrasekharDynamicalFrictionForce` agrees to 3% in pericentre and apocentre (1.84/47.1 versus 1.79/46.0 kpc; 1 s versus 482 s).

4.2 Assumptions, and which one matters

Friction scales with the satellite’s bound mass, so the backward orbit depends on the *function* $M(t)$. Integrating with a constant mass is not a conservative simplification: it is wrong in the direction that matters. A constant $10^{10} M_{\odot}$ satellite is pumped to 40–140 kpc within 10 Gyr, i.e. it “arrives” from outside the virial radius although the real object had already been stripped to a small fraction of that mass for most of its orbit (Fig. 2, right).

We therefore parametrise the history and let a physical requirement select it:

1. **Infall mass** M_{inf} . The nuclear-cluster–host relation gives $M_{\star} \approx 10^9 M_{\odot}$ for the host (Pfeffer et al., 2021); oMEGACat X quotes a “GE-like” dwarf mass $4.5^{+5.7}_{-2.4} \times 10^9 M_{\odot}$ and favours Sequoia/Thamnos over GSE as the debris (Souza et al., 2026). Halo masses for $M_{\star} = 10^8$ – $10^9 M_{\odot}$ at $z \sim 2$ are 10^{10} – $10^{11} M_{\odot}$; GSE itself is $2 \times 10^{11} M_{\odot}$ (Naidu et al., 2021). Scanned: 10^{10} , 3×10^{10} , 10^{11} , $2 \times 10^{11} M_{\odot}$.
2. **Infall time** $t_{\text{inf}} = 10$ Gyr ago ($z \approx 2$; the GE merger is dated 8–11 Gyr ago). Also scanned 8 Gyr; same conclusions.
3. **Stripping law** $M(t) = M_{\text{inf}} \exp[-(t_{\text{inf}} - t_{\text{lb}})/\tau]$, floored at the nucleus mass $3.55 \times 10^6 M_{\odot}$ (Baumgardt & Hilker, 2018); τ scanned 0.5–3 Gyr. Half-mass radius $r_{\text{h}} = 1$ kpc $(M_{\text{inf}}/10^{10} M_{\odot})^{1/3}$, fixed.
4. **Selection rule**. The orbit must be at the virial radius of the young Milky Way, 50–150 kpc, at t_{inf} . Histories that do not reach 50 kpc could not have sunk to today’s orbit by friction alone; histories beyond 150 kpc would need more friction than the mass allows.

4.3 Scan and picks

Figure 3 shows the apocentre in the last Gyr before infall over the (M_{inf}, τ) grid. The table is noisy at the 20% level because the phase of the last apocentre matters, but the trends are clear:

a Sequoia-scale $10^{10} M_{\odot}$ halo reaches the virial radius only with slow stripping ($\tau \gtrsim 1.5$ Gyr); a GSE-scale halo must be stripped fast ($\tau \lesssim 1$ Gyr) or it arrives from far outside the virial radius. Three histories spanning the allowed band are kept (Table 3). In every viable history the bound mass is below $10^9 M_{\odot}$ by 5 Gyr ago and the orbit has today’s shape for the last 3–5 Gyr (Fig. 2, middle): *the present orbit is a good proxy for the second half of the history in all cases*, and the classes differ in the first half.

Table 3: Class-2 picks (kpc). Columns 5–6: pericentre–apocentre range in the stated look-back window.

label	$M_{\text{inf}} [M_{\odot}]$	τ [Gyr]	r at infall	5–6 Gyr ago	2–3 Gyr ago
Sequoia-like	10^{10}	2.0	65	2.9–17.9	2.0–9.2
intermediate	3×10^{10}	1.25	85	2.2–11.6	1.7–7.4
GSE-like	10^{11}	0.75	111	1.7–7.5	1.6–7.1

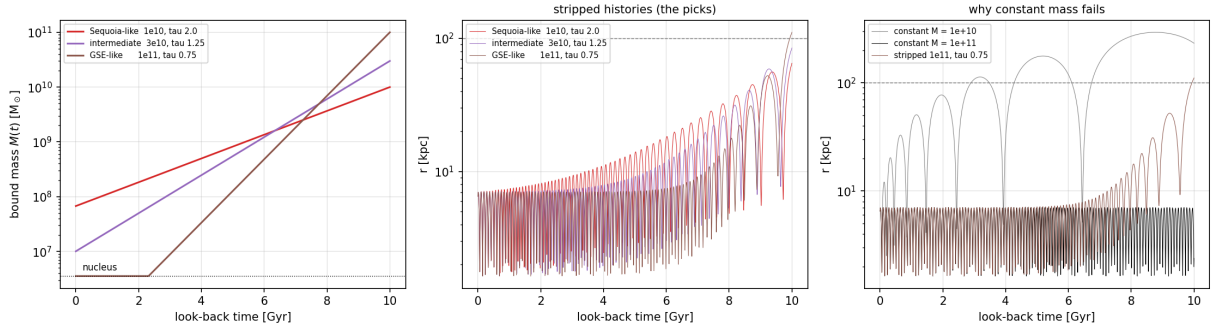


Figure 2: Class 2. Left: the three adopted bound-mass histories. Middle: the corresponding backward orbits; the dashed line marks 100 kpc. Right: why a constant mass fails — a constant $10^{10} M_{\odot}$ satellite (grey) is pumped beyond the virial radius within 10 Gyr, a constant $10^{11} M_{\odot}$ one (black) escapes even faster, while the stripped 10^{11} history (brown) stays on today’s orbit for 6 Gyr and arrives from 110 kpc.

5 Class 3: deposited by GSE, migrated inward by the bar

5.1 What the literature provides

Dillamore et al. (2026) showed that ω Cen can be traced back to the GSE debris region of (L_z, E) if the Galactic bar has decelerated and its present pattern speed is $\Omega_b \lesssim 26 \text{ km s}^{-1} \text{ kpc}^{-1}$, well below most current estimates. The mechanism is the retrograde 1:1 resonance, $\Omega_\phi - \Omega_b + \Omega_r = 0$, which scatters orbits to lower E and more retrograde L_z . Friction on the nucleus is negligible ($\dot{L}_z \approx 9.6 \text{ kpc km s}^{-1} \text{ Gyr}^{-1}$). The GSE debris itself is characterised by Belokurov et al. (2023): $|L_z| < 700 \text{ kpc km s}^{-1}$, chevrons with apocentres 11.5–25 kpc, and a set of (L_z, E) contours that we use directly.

5.2 Construction: reproducing the published set-up

The authors’ pipeline (repository `adllmr/oCen_bar`, local clone `~/Work/Code/oCen_bar`) was followed step by step; its potential files and contour file are referenced, not copied.

- **Potential.** Hunter et al. (2024) Milky Way with the Sormani et al. (2022) bar ($M_{\text{bar}} = 1.83 \times 10^{10} M_{\odot}$), as AGAMA files. The model is decomposed as $\Phi = \Phi_{\text{axi}} + A(t) \Phi_{\text{bar}}[S(t)] -$

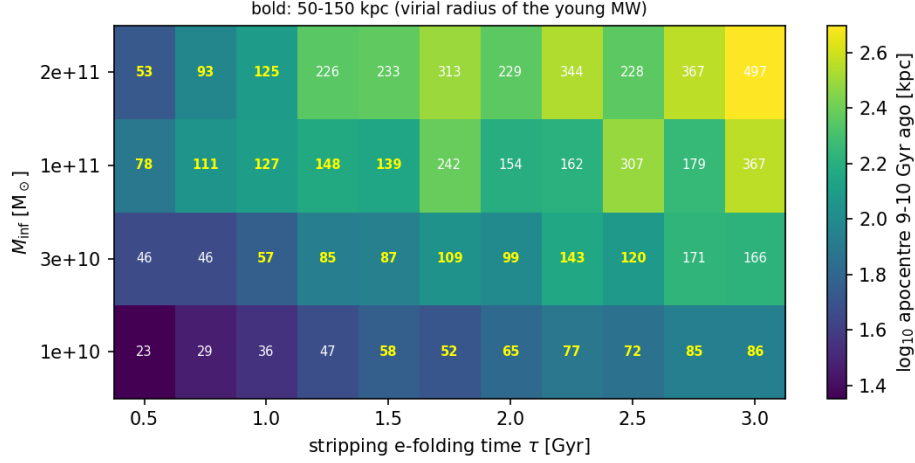


Figure 3: Class 2: apocentre 9–10 Gyr ago as a function of infall mass and stripping e-folding time ($t_{\text{inf}} = 10$ Gyr). Bold entries satisfy the 50–150 kpc selection rule. Low-mass halos need slow stripping, GSE-mass halos fast stripping.

$A(t) \bar{\Phi}_{\text{bar}}[S(t)]$, i.e. the axisymmetric total plus the scaled, rotating baryonic bar part minus the same part azimuthally averaged, so the $m = 0$ mass is unchanged while the bar grows.

- **Bar history.** Amplitude A follows the [Dehnen \(2000\)](#) switch-on between $t = 0$ and $t_1 = 1$ Gyr. The pattern speed is Ω_1 until t_1 , starts decelerating smoothly to $t_2 = 2$ Gyr, then obeys $\eta \equiv -\dot{\Omega}_b/\Omega_b^2 = 0.003$ until $t_f = 8$ Gyr (the bar age). Ω_1 is fixed by the present-day Ω_b : $\Omega_1 = 45.1 \text{ km s}^{-1} \text{ kpc}^{-1}$ for $\Omega_b = 24$ (the paper quotes “ ≈ 45 ”, which confirms $t_1, t_2 = 1, 2$ Gyr rather than the 2, 3 defaults left in one of the repository functions), 72 for 30, 110 for 35. The bar length scales as $S(t) = \Omega_b(t_f)/\Omega_b(t)$, approximately the corotation radius, at fixed amplitude (Fig. 6, left).
- **Samples.** $N = 10^3$ draws of the present-day state (Sec. 2), bar angle 28° , integrated from t_f back to 0 with `agama.orbit`.
- **Test.** (E, L_z) at $t = 0$ in the axisymmetric potential, compared with the outermost [Belokurov et al. \(2023\)](#) contour after matching the energy zero point at $R = 8.2 \text{ kpc}$. Our success metric is the fraction of samples inside that contour.

5.3 Results

Table 4: Fraction of the 10^3 samples inside the outer GSE contour 8 Gyr ago.

$\Omega_b \text{ [km s}^{-1} \text{ kpc}^{-1}]$	20	21	22	22.5	23	23.5	24	24.5	25	25.5	26	26.5	≥ 27
fraction	0.44	0.38	0.58	0.85	0.01	0.34	0.89	0.81	0.79	0.54	0.18	0.02	0.00

- The published result is reproduced: overlap with GSE only for $\Omega_b \lesssim 26$, none for the mainstream $30\text{--}40 \text{ km s}^{-1} \text{ kpc}^{-1}$ (Fig. 4).
- Cross-check against the authors’ stored grid (`oCen_bar/artifacts/03_orbit_grid.npz`): 0.87, 0.69, 0.08 at $\Omega_b = 24, 25, 26$ with a dip at 23 — the same pattern. The dip is stable against the random seed (0.013 versus 0.028) and against the bar timings: it is resonant-phase sensitivity, not noise. The Ω_b dependence is intrinsically jagged.

- **Migration is late.** E and L_z of the successful samples stay at GSE-like values from 8 to ~ 2.5 Gyr ago and move to today’s values only in the last ~ 2.5 Gyr, while Ω_b falls from ~ 30 to 24 and the resonance sweeps across ω Cen (Fig. 5). The pre-migration L_z is *slightly prograde to zero* (+70 to +230 kpc km s $^{-1}$), not retrograde.
- **The bar changes today’s orbit.** At $\Omega_b = 24$ corotation is at ≈ 9.5 kpc, so ω Cen at 6.5 kpc is inside corotation. Its present orbit in the barred potential has peri/apo = 0.81/9.3 kpc over the last Gyr, versus 1.56/7.02 in the axisymmetrised Hunter potential and 1.57/7.04 in McMillan 2017. “Today’s orbit” in class 1 is therefore conditional on the bar being fast or absent.

Table 5: Class-3 picks at $\Omega_b = 24$: samples at the 16/50/84th percentiles of $E(t = 0)$ among those ending inside GSE. Elements from a 0.25-Myr re-integration (the 20-Myr grid cadence overestimates pericentres by up to a factor 2, since a passage at 0.4 kpc lasts ~ 1 Myr).

sample	$E(0)$ [10^5]	$L_z(0)$	pre-migration peri/apo/ $e/z_{\max}$ (7–8 Gyr ago)	last Gyr peri/apo
986	−1.301	+69	0.40 / 10.9 / 0.93 / 4.8	0.84 / 9.1
133	−1.265	+163	0.42 / 11.6 / 0.93 / 4.9	0.80 / 9.3
240	−1.233	+229	0.45 / 12.2 / 0.93 / 5.2	0.84 / 9.1

5.4 What is still not modelled in class 3

Anything before the bar formed (> 8 Gyr ago): GSE’s own infall and disruption (fiducial: $M_\star = 5 \times 10^8$, $M_{\text{DM}} = 2 \times 10^{11} M_\odot$, $c = 4$, from R_{vir} at $z \approx 2$, circularity 0.5, retrograde; Naidu et al., 2021) and where ω Cen’s host sat in it. Two readings of “deposited by GSE” remain open — ω Cen is GSE’s own nucleus (Pfeffer et al., 2021), or its host was a smaller dwarf that fell in with GSE (group infall, the oMEGACat X reading) — and both start the disruption on the picked debris orbit. The bar history is a one-parameter family at fixed η ; the oCen_bar journal notes that the low- Ω_b requirement is tied to this slowdown law, since $\Omega_b = 35$ today implies an implausible $\Omega_1 = 110 \text{ km s}^{-1} \text{ kpc}^{-1}$.

6 Comparison between the classes

Figure 8 and Table 6 summarise what a disruption simulation would see.

Table 6: Per-Gyr elements at three epochs (kpc unless stated). Tidal strength $T \equiv GM(< r_{\text{peri}})/r_{\text{peri}}^3$ in $10^3 (\text{km s}^{-1}/\text{kpc})^2$; r_J is the Jacobi radius of a $10^9 M_\odot$ host at pericentre in the last Gyr.

class	orbit	peri (0–1 / 5–6 / 8–9 Gyr)	apo (0–1 / 5–6 / 8–9)	T (0–1)	T (8–9)	$r_J^{10^9}$
1	McMillan 2017	1.59 / 1.58 / 1.58	7.0 / 7.0 / 7.0	14.9	14.9	0.46
1	MWPotential2014	1.98 / 1.98 / 1.97	6.8 / 6.8 / 6.8	10.3	10.3	0.52
1	Irrgang 2013 I	1.25 / 1.26 / 1.25	7.2 / 7.2 / 7.2	21.7	21.7	0.40
2	Sequoia-like	1.65 / 2.91 / 4.84	7.4 / 17.9 / 45	13.9	1.9	0.47
2	intermediate	1.64 / 2.24 / 5.99	7.1 / 11.6 / 41	14.1	1.4	0.47
2	GSE-like	1.64 / 1.72 / 5.01	7.0 / 7.5 / 33	14.1	1.9	0.47
3	sample 986	0.84 / 0.59 / 0.40	9.1 / 11.0 / 10.9	39.4	104	0.33
3	sample 133	0.80 / 0.71 / 0.42	9.3 / 11.7 / 11.6	42.0	100	0.32
3	sample 240	0.84 / 0.80 / 0.45	9.1 / 11.1 / 12.2	39.4	92	0.33

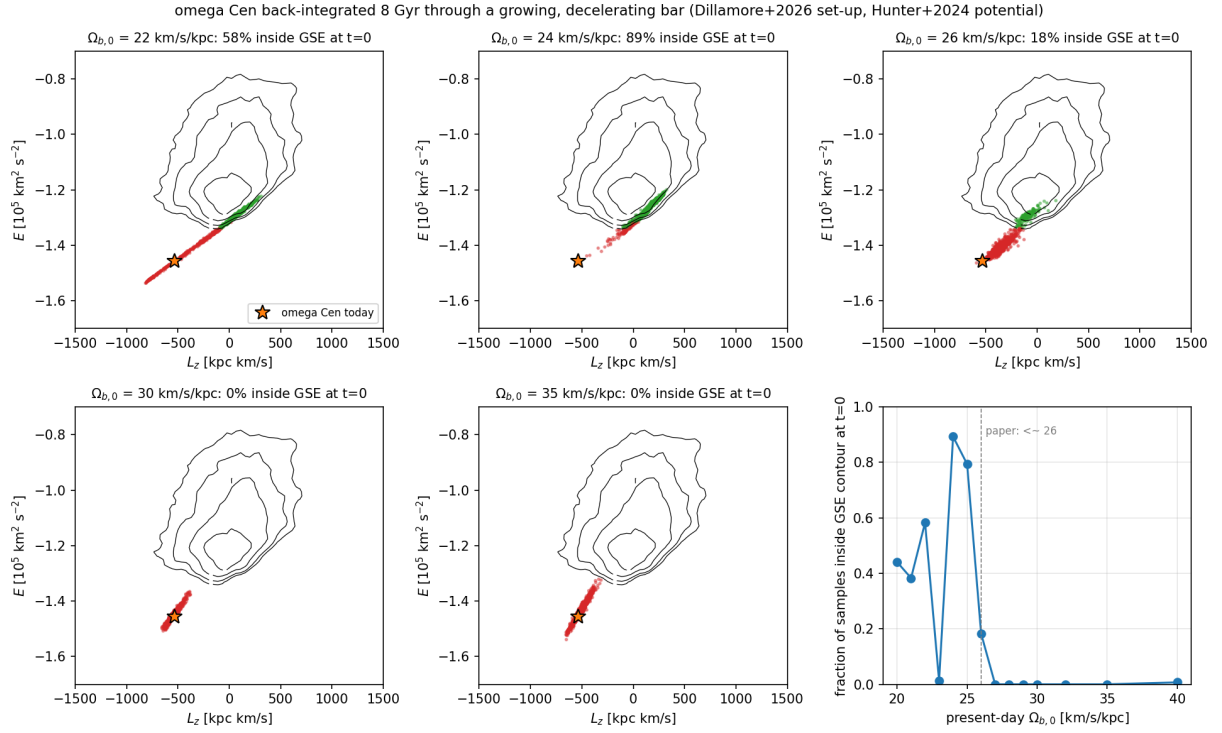


Figure 4: Class 3, the (L_z, E) plane in the axisymmetrised Hunter et al. (2024) potential. Black: GSE contours of Belokurov et al. (2023). Star: ω Cen today. Points: the 10^3 samples 8 Gyr ago after back-integration through the decelerating bar, green inside / red outside the outer contour. Bottom right: fraction inside versus present-day pattern speed; the dashed line is the paper's $\Omega_b \lesssim 26$.

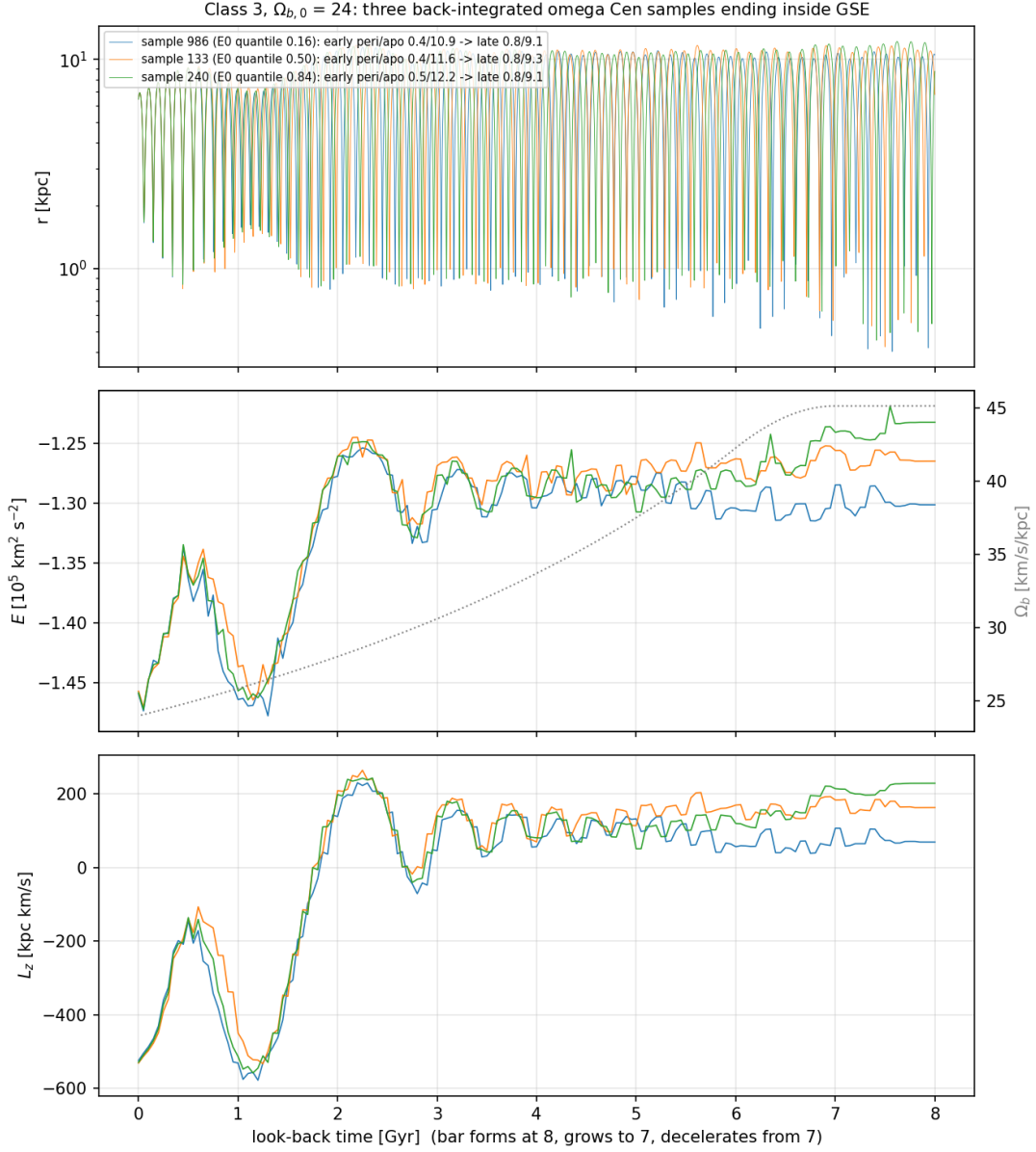


Figure 5: Class 3, the three picked samples at $\Omega_b = 24$: radius, energy (dotted grey: $\Omega_b(t)$) and L_z against look-back time. The bar forms 8 Gyr ago and grows until 7; the nucleus sits on a GSE-debris orbit until ~ 2.5 Gyr ago and is then carried to today's (E, L_z) by the retrograde 1:1 resonance. Look-back time here is $t_f - t$.

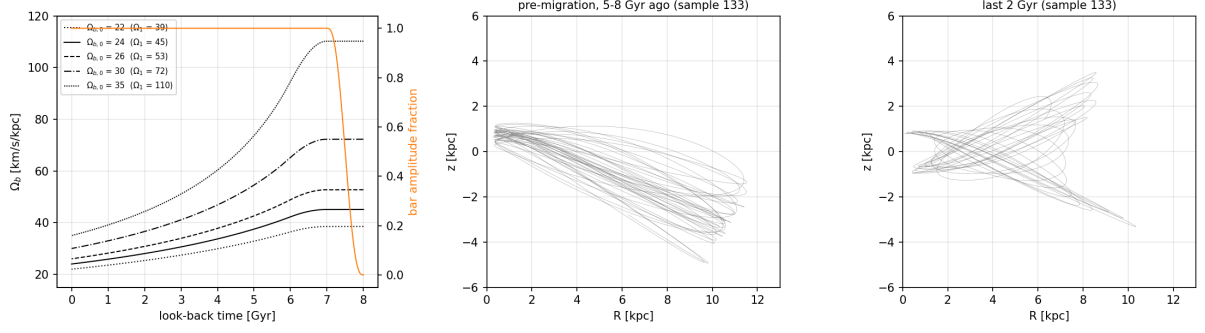


Figure 6: Class 3. Left: pattern-speed histories for several present-day values (black) and the bar amplitude switch-on (orange, right axis); with fixed $\eta = 0.003$ a present-day $\Omega_b = 35$ demands an initial $\Omega_1 = 110 \text{ km s}^{-1} \text{ kpc}^{-1}$. Middle and right: the median pick (sample 133) in the meridional plane before migration and in the last 2 Gyr — a plunging, highly inclined debris orbit turning into a rounder, lower orbit.

Late history (last 3 Gyr). Classes 1 and 2 are indistinguishable: pericentre 1.6 kpc, apocentre 7 kpc, $T \approx 1.4 \times 10^4 (\text{km s}^{-1}/\text{kpc})^2$. Class 3 is different even today because the slow bar is part of its potential: pericentre 0.8 kpc and three times stronger pericentric tides. Whether that is a property of class 3 or of the Galaxy depends only on whether the bar is slow; if it is, classes 1 and 2 should be re-evaluated in the barred potential as well.

Early history (5–10 Gyr ago). This is where the classes separate, by more than an order of magnitude in tidal strength. Class 1 keeps today’s tides throughout. Class 2 has a gentle early history: pericentres of 2–6 kpc and T ten times weaker than today, so the host is stripped slowly at first and hardest at the end, when it is already light. Class 3 is the opposite: pericentres of 0.4 kpc through the bulge with T seven times *stronger* than today and a Jacobi radius of only 0.3 kpc for a $10^9 M_\odot$ host — the host is destroyed early and violently, and what survives to be migrated by the bar is the nucleus alone. For the dark-matter question this is the sharpest distinction: a class-3 history leaves the nucleus the least time to retain a dark envelope, a class-2 history the most.

Angular momentum. The early orbit is more retrograde than today in class 2 ($L_z \rightarrow -2000$ to $-4000 \text{ kpc km s}^{-1}$ as friction is undone) and *less* retrograde in class 3 ($L_z \approx 0$ to $+230$; Fig. 9). Retrograde class-2 infall is consistent with the retrograde Sequoia/Thamnos association; a class-3 host arrives on the near-radial GSE orbit.

Robustness. Class 1 depends only on the potential (factor 1.6 in pericentre). Class 2 depends on $M(t)$ far more than on any single mass, and the selection rule of Sec. 4.2 is what makes the set finite; the friction formula itself is a 20–30% approximation. Class 3 depends on the bar’s slowdown law and on a present-day pattern speed at the low end of the literature, and its success is a jagged function of Ω_b .

7 Caveats and reproducibility

- Chandrasekhar friction with $\sigma = v_c/\sqrt{2}$, a fixed r_h and a smooth exponential stripping law; no orbit-phase-dependent mass loss.
- Only McMillan (2017) for class 2; only Hunter et al. (2024) for class 3 (the only potential with a fitted bar); class 1 carries the potential spread.

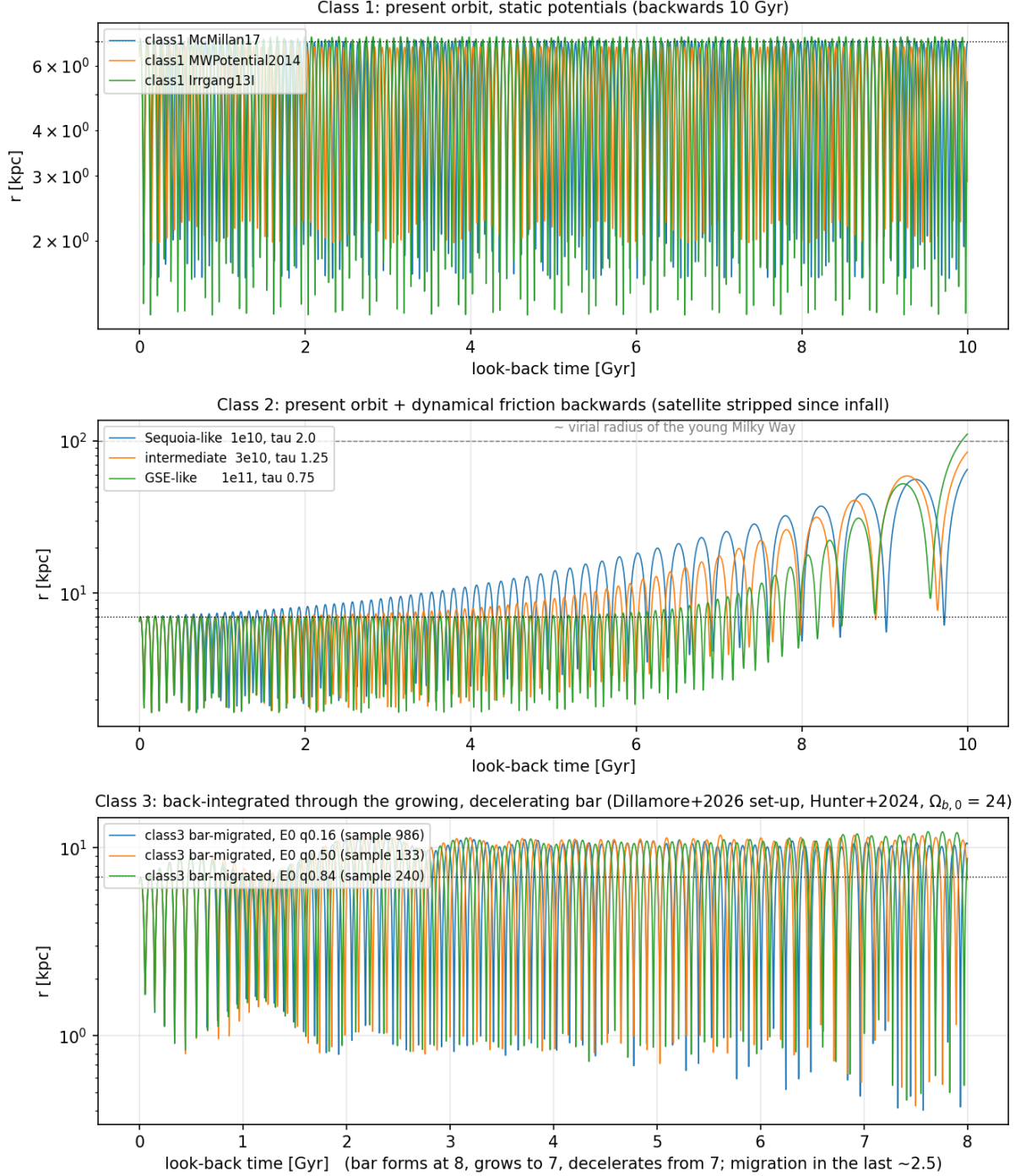


Figure 7: Overview of the nine orbits: galactocentric radius against look-back time for classes 1 (top), 2 (middle; dashed line 100 kpc) and 3 (bottom). Dotted line: today's apocentre, 7 kpc.

- Tidal proxies in Fig. 8 use McMillan (2017) enclosed masses for all classes.
- The `galpy` checkout used (2024-03) needs two import shims (`astroquery metadata`, `scipy vectorize1`); both live in `progenitor_orbits._import_galpy_safely`.
- Products: `python -m ocen_dm.tails.products`, `python -m ocen_dm.tails.bar_migration`, `python -m ocen_dm.tails.writeup_figures`. The class-3 run of 14×10^3 orbits takes about a minute.

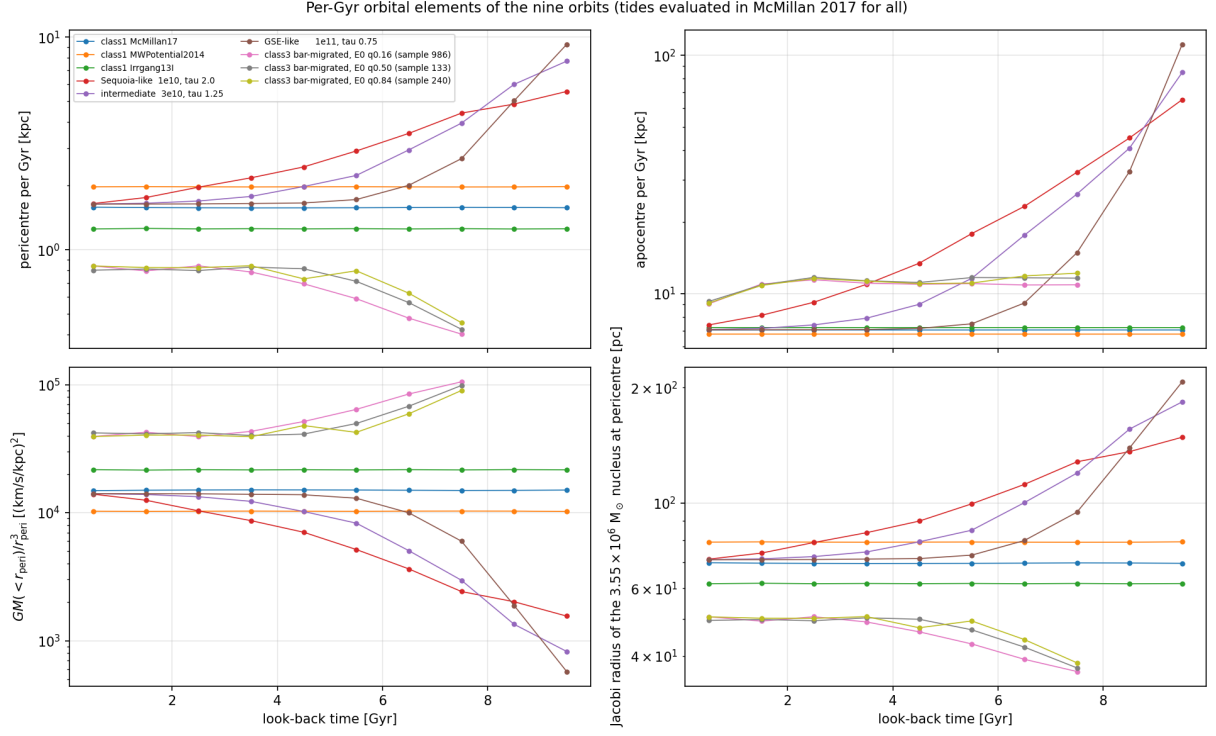


Figure 8: Per-Gyr orbital elements of the nine orbits: pericentre, apocentre, tidal-field strength $GM(< r_{\text{peri}})/r_{\text{peri}}^3$ at pericentre, and the Jacobi radius a $3.55 \times 10^6 M_{\odot}$ nucleus would have there. Enclosed masses from McMillan (2017) for all orbits, including class 3 (integrated in Hunter et al. 2024), for a uniform proxy.

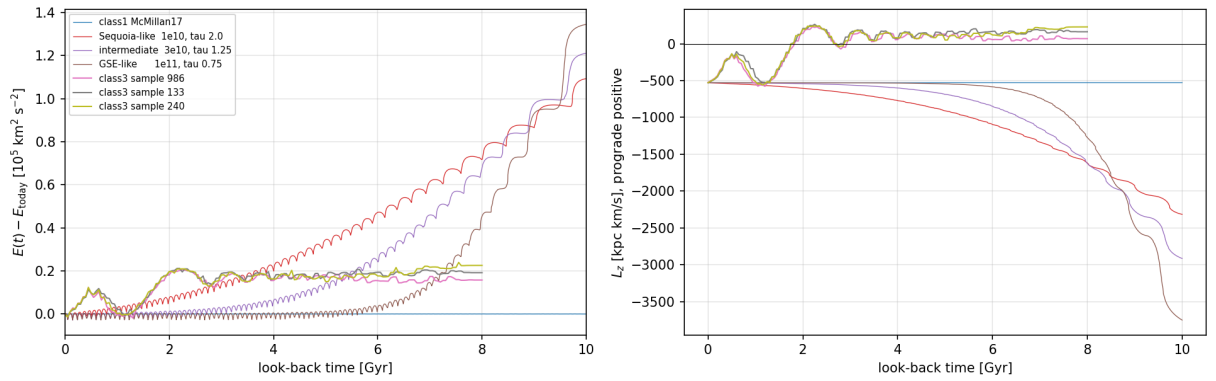


Figure 9: Energy relative to today (left; each orbit in its own host potential) and L_z (right, prograde-positive) against look-back time. Class 1 (McMillan 2017 only) is flat by construction; class 2 climbs steadily as friction is undone; class 3 is flat at GSE values until the resonance moves it in the last 2.5 Gyr. Note the opposite directions of the L_z change: friction makes the early orbit *more* retrograde, the bar makes it *less*.

References

- Baumgardt, H., Hilker, M. 2018, MNRAS, 478, 1520, <https://doi.org/10.1093/mnras/sty1057>
- Baumgardt, H., Vasiliev, E. 2021, MNRAS, 505, 5957, <https://doi.org/10.1093/mnras/stab1474>
- Belokurov, V., Vasiliev, E., Deason, A. J., et al. 2023, MNRAS, 518, 6200, <https://arxiv.org/abs/2208.11135>
- Bovy, J. 2015, ApJS, 216, 29, <https://doi.org/10.1088/0067-0049/216/2/29>
- Chandrasekhar, S. 1943, ApJ, 97, 255, <https://doi.org/10.1086/144517>
- Dehnen, W. 2000, AJ, 119, 800, <https://doi.org/10.1086/301226>
- Dillamore, A. M., Zhang, H., Belokurov, V. 2026, arXiv:2606.12516, <https://arxiv.org/abs/2606.12516>
- Hunter, G. H., Sormani, M. C., Beckmann, J. P., et al. 2024, A&A, 692, A216, <https://www.aanda.org/articles/aa/pdf/2024/12/aa50000-24.pdf>
- Irrgang, A., Wilcox, B., Tucker, E., Schiefelbein, L. 2013, A&A, 549, A137, <https://doi.org/10.1051/0004-6361/201220540>
- McMillan, P. J. 2017, MNRAS, 465, 76, <https://doi.org/10.1093/mnras/stw2759>
- Naidu, R. P., Conroy, C., Bonaca, A., et al. 2021, ApJ, 923, 92, <https://arxiv.org/abs/2103.03251>
- Pfeffer, J., Lardo, C., Bastian, N., Saracino, S., Kamann, S. 2021, MNRAS, 500, 2514, <https://academic.oup.com/mnras/article/500/2/2514/5956542>
- Schönrich, R., Binney, J., Dehnen, W. 2010, MNRAS, 403, 1829, <https://doi.org/10.1111/j.1365-2966.2010.16253.x>
- Sormani, M. C., Gerhard, O., Portail, M., Vasiliev, E., Clarke, J. 2022, MNRAS, 514, L1, <https://doi.org/10.1093/mnrasl/slac046>
- Souza, S. O., Neumayer, N., Seth, A. C., et al. 2026, arXiv:2603.23589, <https://arxiv.org/abs/2603.23589>
- Vasiliev, E. 2019, MNRAS, 482, 1525, <https://doi.org/10.1093/mnras/sty2672>
- Vasiliev, E., Baumgardt, H. 2021, MNRAS, 505, 5978, <https://doi.org/10.1093/mnras/stab1475>